# Lost & Found Network

## Project Proposal Report

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Group 84
Akinwunmi Eludoyin – 101567797
Blossom Babalola – 101606071
Derek Page – 101590623
Seyma Sibel Ekinci – 101323633
Sofia Janik – 101573681

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# 1. Project Context and Client Focus

## 1.1 Project Background

George Brown Polytechnic operates across multiple campus locations, each with numerous buildings, classrooms, common areas, and shared facilities where personal belongings are frequently misplaced. Currently, lost and found services are fragmented across security desks, departmental offices, and library service points, with no unified system for tracking or publicizing found items. As a result, individuals who lose belongings often do not know where to inquire, leading to delayed recovery, unclaimed items, and frustration within the campus community.

The **Lost & Found Network** is proposed as a centralized, web-based platform designed to address this gap. The system will allow members of the college community to report lost and found items in one place, search existing listings, and securely claim belongings. Primary users include students who may lose personal items such as phones, wallets, or textbooks; faculty and staff who encounter or lose items in offices and shared spaces; and campus visitors attending events or using facilities such as the library or gym. By consolidating information and standardizing the process, the platform aims to improve recovery rates, reduce administrative overhead, and enhance the overall campus experience.

## 1.2 Client or Stakeholder Focus

The primary clients and stakeholders for the Lost & Found Network include students, faculty and staff, and Lost & Found or Security staff. These groups interact directly with the system and are most impacted by its effectiveness and usability.

| Stakeholder | Role | Interests / Goals | Influence |
|---|---|---|---|
| **Students** | Primary users who lose or find items on campus | Quickly report lost items, search for found items, securely recover belongings | High |
| **Faculty and Staff** | Users and occasional administrators of lost items | Efficient handling of lost property, reduced manual work, fewer inquiries | High |
| **Campus Visitors** | Occasional users (guests, event attendees) | Simple way to report and recover lost belongings | Low |

| | | | |
|---|---|---|---|
| **Lost & Found Office / Security Staff** | Operational managers of the process | Accurate tracking, dispute resolution, secure item handover | High |
| **IT Department** | System maintenance and support | System reliability, security, scalability | Medium |
| **College Administration** | Institutional oversight | Improved service quality, reduced complaints, accountability | Medium |

Students, faculty, and staff represent the most critical stakeholder groups, as they are the primary users of the system and experience the greatest impact from current inefficiencies. Lost & Found and Security staff have high influence because they are the ones who check ownership, deal with disputes, and return items.

While the IT department and college administration are not daily users of the system, they play a key role in ensuring data security, system reliability, and long-term sustainability. The system must therefore balance ease of use for end users with appropriate controls and oversight for staff and administrators.

# 2. Problem Statement and Business Need

## 2.1 Business Need

A centralized lost and found system is needed to reduce the administrative burden on campus staff and improve the recovery rate of lost items. Lost item owners need a single place to search for their belongings and receive notifications about potential matches. Individuals who find items need a simple way to report them and connect with owners. Lost & Found Office staff need visibility into all reported items and a verification workflow to ensure secure returns and resolve disputes.

## 2.2 Problem Statement

Students, faculty, and staff currently search for lost items by contacting multiple disconnected offices across campus, which leads to wasted time, low recovery rates, and frustration. Lost items may be turned in to any of several locations including security desks, departmental offices, and library service points. There is no centralized system that allows individuals to report lost or found items, search listings, and submit verified claims in one

place, resulting in unclaimed belongings, financial loss for students, and increased administrative workload for staff.

# 3. Project Objectives and Scope

## 3.1 Project Objectives

- Create a website that enables users to report lost and found items
- Implement user authentication so students, staff and visitors can safely create and manage reports
- Provide search and filtering to improve the efficiency and success rate of item recovery
- Create a claim verification process that will return the items to their owners
- Deliver a fully functional system by the end of the semester

## 3.2 Scope – In Scope

- Account creation using email, login and logout
- Creating lost item reports with category, description, date lost, location, and optional image
- Submission of reports for found items with item details
- Search and browse functionality with filters for category, date time, location, and status
- Claim submission, approval, and rejection process
- Admin dashboard for managing listings
- Email notifications for potential matches or approved claims
- Web interface accessible on modern desktop and mobile browsers

## 3.3 Scope – Out of Scope

- Mobile application
- Real-time chat or direct messaging between users
- Image recognition for lost and found items
- SMS notifications
- Physical item storage tracking
- Advanced analytics dashboards
- Automatic content moderation

# 4. Proposed Solution and System Overview

## 4.1 System Type and Users

The Lost & Found Network will be developed as a web-based application that users can access through a modern web browser on desktop or mobile devices. The system will not require any installation, which makes it easier for students, staff, and visitors to use anytime and from anywhere on campus.

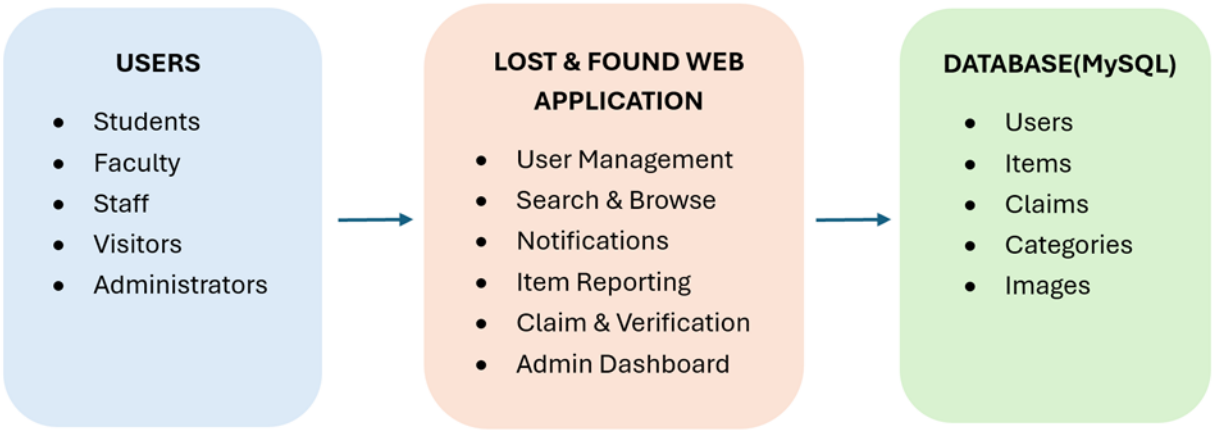
The main users of the system include:

- **Students** report lost items, search listings, submit claims, receive notifications
- **Faculty and staff** report, recover, or manage items
- **Campus visitors** can register with an email address to report or recover items during their campus visit.
- **Lost & Found Office/Administrators** review claims, manage listings, handle disputes, and generate reports.
- **IT Department** monitors system performance and manages backups.

## 4.2 System Overview

The Lost & Found Network is a centralized web-based platform that consolidates all lost and found activities into one digital system. Users interact with the application through a web interface, which communicates with a backend server and MySQL database to manage all data and operations.

The system includes modules for User Management, Search & Browse, Notifications, Item Reporting, Claim & Verification, and Admin Dashboard.



## 5. Technology Stack and Tools

| Layer | Technology | Justification |
|---|---|---|
| **Frontend** | JavaScript (Language) | This will be used for both the frontend and backend of the application. Our team has already worked with JavaScript in previous semesters, so we are comfortable with the syntax. Using one language across the entire project also helps keep the code more consistent and easier to maintain, since we don't have to switch between different programming languages for client and server logic |
| | React (Framework) | This will be used to build the frontend user interface. It is well suited for creating dynamic pages such as item listings, search filters, and dashboards. React's component-based structure makes it easier to reuse UI elements and keep the project organized as it grows. Our frontend lead has great interest in React which will positively impact development speed |
| **Backend** | JavaScript (Languages) | Used on frontend, keeping language across stack consistent |
| | Nodejs (Library) | Will handle the backend server logic and communication with the database. Since it also uses JavaScript, it works smoothly with React and allows some logic and |

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|---|---|---|
| | | validation to be shared between frontend and backend if needed. Node.js is also known for handling multiple requests efficiently, which is useful for a system where many users may be browsing or submitting reports at the same time. Our backend lead already has strong experience with Node.js, which will make for a smoother and more reliable development |
| **Database** | MYSQL | This will be used as the database management system. Our team has experience in MySQL from previous semesters, which makes it the most practical and low risk choice. MySQL works well for structured data and relationships which fits this project since we need to connect users, items, and claims. It is also widely used, reliable, and supports backup and replication features if the system ever needs to scale |
| **Version control & Project Management** | Git / GitHub | This will be used for version control and project management. Our team has used GitHub in past courses and is currently taking a class that has a focus on GitHub. GitHub allows multiple team members to work at the same time using branches, helps track changes through commits, and |

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| | | provides strong tools for resolving conflicts |
| **Others** | 3rd APIs or services | As the project grows, the team could employ deployment tools or external auth validation such as JWT or existing libraries to make development and code execution more efficient. |

# 6. Feasibility, Constraints, and Risks

## 6.1 Feasibility Summary

The Lost & Found Network for Campus project is feasible within the COMP 2154 semester timeline because the scope has been limited to essential core features, including authentication, item reporting, search functionality, and claim management. The system does not attempt to integrate with external institutional systems or implement advanced automation, which helps keep development realistic.

The team's technical skills align with the requirements of a web-based application involving front-end development, back-end logic, and database management. By following an Agile, week-by-week approach with clearly defined milestones, the team can incrementally develop, test, and refine the system throughout the semester.

## 6.2 Constraints

- **Time Constraint:** The project must be completed within the academic semester, limiting the scope to core functionality.
- **Data Constraint:** Real campus lost and found data will not be used; testing will rely on simulated or sample data.
- **Security & Privacy Constraint:** User information must be handled responsibly, even in a prototype environment.
- **Resource Constraint:** Development is limited to team availability and academic workload.

## 6.3 Initial Risks and Mitigations

- **Scope Creep:** Adding extra features beyond core requirements may delay completion.

*Mitigation:* Strictly prioritize backlog items and focus on minimum viable functionality first.

- **Technical Challenges:** Integration between front-end, back-end, and database components may take longer than expected.

*Mitigation:* Begin integration early and test features incrementally.

- **Team Coordination Risk:** Miscommunication or uneven workload distribution could affect progress.

*Mitigation:* Maintain weekly meetings, clear task ownership, and use GitHub for tracking issues.

- **Time Management Risk:** Academic workload from other courses may reduce available development time.

*Mitigation:* Follow the weekly schedule and complete milestones progressively rather than delaying work.

# 7. Project Plan and Schedule

## 7.1 Approach: Gantt Chart or Agile Backlog

The project follows a standard Agile, iteration-based approach organized by weekly milestones. Development work will be tracked using a shared Kanban-style backlog on Github platform, with user stories derived directly from the functional requirements defined in the requirement report. Each week focuses on delivering an agreed set of features while allowing for continuous review, testing, and refinement.

## 7.2 Major Phases and Milestones

The project is divided into the following high-level phases:

- **Planning & Design (Weeks 4–5):** Finalize requirements, tech stack, project roles, project planning and coordination tools.
- **Core Feature Implementation (Weeks 6–10):** Database modelling, UI wireframes and development, authentication, reporting, browsing, and administrative features
- **Stabilization & Deployment (Weeks 11–12):** Deploy & Preview MVP, refine UX, address QA concerns
- **Final Review & Delivery (Weeks 13–15):** Prepare presentation materials, conduct demo rehearsals, and submit final deliverables

## 7.3 Task Allocation

Responsibilities are shared across the team, with technical leads coordinating backend and frontend implementation, and a project lead overseeing documentation, planning, and presentation readiness.

| Week | Focus | Key Deliverables / Milestones | Owner(s) |
|---|---|---|---|
| **3** | Project Proposal | Assign and review project proposal | All |
| **4** | Project Proposal completion | Review and complete project proposal | All / Documentation lead |
| **5** | Architectural & setup | Repo setup, DB schema draft, core entities defined, UI wireframes, finalized user stories | Tech leads |
| **6** | Core Features – Auth & Reporting | User authentication, role definitions (user/admin), lost & found item reporting (FR-1 to FR-4) | Frontend / Backend |
| **7** | Core Features – Browsing & Search | | |
| **(Midterm week)** | Browse listings, filtering by category/location/status (FR-5, FR-7) | All | |
| **8** | Study week | Buffer, refactor, technical debt cleanup, midterm feedback incorporation | All |
| **9** | Core Features – Claim Flow | Claim submission, verification flow, status updates; midterm demo-ready MVP | All |

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|---|---|---|---|
| **10** | Admin & Notifications | Admin listing management, dispute handling, email notifications (FR-9 to FR-12) | All |
| **11** | Deployment planning | Deploy MVP, environment configuration, backup & recovery notes | DevOps Lead |
| **12** | Refinement & UX | UX improvements, accessibility checks, performance polish | All / Tech Lead |
| **13** | Presentation prep | Slides, demo script, walkthrough scenarios, documentation review | Project lead |
| **14** | Review, refine | Full demo dry run, final fixes, report finalization | All |
| **15** | Presentation Execution | Final report submission and system demo/presentation | All |

Planned link to project management tool: Project link

# 8. Team Structure and Communication Plan

## 8.1 Team Roles

The project team is organized with clearly defined roles to ensure accountability, efficient collaboration, and timely delivery. While responsibilities are assigned by role, team members may support multiple areas as needed.

- **Scrum Lead:** Facilitates weekly check-ins, tracks progress and blockers via the kanban board, and coordinates task flow across the team.
- **Documentation Lead:** Manages proposal and report formatting, ensures alignment with course requirements, and coordinates on-time submissions.

- **Frontend Lead:** Designs UI wireframes, defines and assigns user stories, implements frontend features, and conducts code reviews for UI-related work.
- **Backend Lead:** Designs database schema, implements API endpoints, manages business logic, and oversees system integrations.
- **QA Lead:** Develops test plans, executes end-to-end testing, and coordinates defect tracking and resolution.
- **Developers:** Implement assigned features, write unit tests, and participate in code reviews and debugging.

## 8.2 Communication and Collaboration

The team continues to use a combination of synchronous and asynchronous communication tools to ensure quick turnaround, alignment, and steady progress throughout the semester.

- **Primary communication:** A shared WhatsApp group chat used for day-to-day communication, quick questions, updates, and coordination.
- **Weekly meetings:** The team holds a weekly check-in every Monday via Microsoft Teams (30–40 minutes). These meetings are used to review progress, discuss blockers, assign upcoming tasks, and align priorities.
- **Task and feature tracking:** Project work will be tracked using GitHub project kanban board. This provides visibility into work status and ownership.
- **Code collaboration:** Git and GitHub will be used for version control, code reviews, and collaboration.

This communication structure ensures that progress is visible, responsibilities are clear, and issues can be addressed promptly.

## 8.3 Decision-Making and Conflict Resolution

The team will make decisions collaboratively, with an emphasis on respect, transparency, and shared accountability.

- **Decision-making process:** Technical and project decisions will be discussed during weekly meetings or asynchronously in the team chat when appropriate. When consensus cannot be reached, decisions will be resolved through a majority vote.
- **Guiding principles:** All decisions and interactions will be guided by mutual respect, open communication, and collaboration, with the goal of selecting solutions that best support project requirements and team success.

This approach ensures that disagreements are handled fairly, efficiently, and without negatively impacting team morale or project progress.

# 9. Assumptions, Dependencies, and Success Criteria

## 9.1 Assumptions

- Students, faculty or campus quests will access the system through a web browser, with either their computer or mobile device
- Information regarding lost items will be manually entered by any user who finds an item
- Both student/guest as well as faculty/administrative roles can be simulated to test functionality
- Features such as item request or owner verification notifications can be tested to ensure functionality

## 9.2 Dependencies

- NodeJS
- ReactJS
- MySQL

## 9.3 Success Criteria

- User and administrators can login through the same portal and access their relevant pages
- Users can login to see information regarding their posted items or requested items
- Administrators can login to manage all item listings or resolve disputes
- Guests to the campus can utilize the system without a George Brown email address
- Items which have been successfully transferred to their owner will automatically be removed from view in user lists and searches.
- Notifications are delivered to users based on their selected preferences
- The system will retain all user account and item listing info in a persistent database
- Depending on time constraints, the system will be fully functional on a deployed platform for demonstration purposes

## 10. Summary and Next Steps

The Lost & Found Network for Campus project focuses on solving the problem of how lost and found items are currently handled at George Brown Polytechnic. Right now, individuals must physically visit a security desk or directly contact security to report or inquire about lost belongings. Although this system functions operationally, it is not digitally centralized or searchable, which limits convenience, transparency, and accessibility for users across multiple campuses.

The main goal of this project is to create a centralized, web-based system where students, staff, and visitors can report lost or found items, browse listings, and submit claims to recover their belongings. The system also aims to make the process more efficient for Lost and Found and Security staff by reducing manual tracking and improving organization.

The proposed solution is a role-based web application that includes item reporting, search and filtering features, claim management workflows, and administrative tools. The project is feasible within the COMP 2154 timeline because the scope is focused on core functionality and aligns with the team's technical skills. Key constraints and risks have been identified and will be managed through careful planning and an iterative Agile approach.

The team will follow a structured, week-by-week development plan aligned with the course schedule to ensure steady progress from system design and implementation through testing and final presentation.

The next steps include finalizing the project proposal, setting up the shared repository and development environment, and defining the initial system architecture and database schema. After setup is complete, development will begin on core features such as user authentication and item reporting. Progress will be tracked through a shared backlog and weekly team meetings to ensure alignment with project milestones.